Gender Peer Effects and Women's Academic Performance in Mathematics: Evidence from a U.S. University

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Abstract: Understanding how educational environments shape women's mathematics achievement and persistence can help explain their continued underrepresentation in mathematics-intensive fields despite rising educational attainment. This paper examines how classroom gender composition affects women's academic performance in introductory mathematics courses at university. Using administrative records for freshmen entering a selective U.S. public university between 2016 and 2023, I exploit variation in female peer share across classes in a specific introductory mathematics course taught by a given instructor within a single academic term, allowing for gender-specific responses. A 10-percentage-point increase in female peer share raises women's probability of completing introductory mathematics by approximately 3.1 percentage points and earning C- or better by approximately 3.4 percentage points. Both women and men experience improvements in broader first-quarter academic progress, with limited evidence of gender differences in the baseline effects. The findings are robust to controlling for gender composition in other courses and excluding pandemic-entry cohorts. Women's benefits are concentrated among students who declared a STEM major at entry, while comparable positive effects do not emerge in other entry-term subjects. Longer-term estimates are consistent with increased STEM degree attainment among STEM entrants, although they remain imprecise. These findings highlight classroom gender composition as a factor supporting women's progress through foundational university mathematics.

Keywords: Gender Peer Effects; Higher Education; Mathematics Achievement; STEM Persistence.

JEL Codes: I21, I23, J16.

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1 Introduction

Despite women's increasing educational attainment, they remain underrepresented in high-earning fields such as engineering and computing (Bertocchi and Bozzano, 2020; Bertrand, 2011). Mathematics preparation and students' perceptions of their mathematical ability can shape participation in mathematics-intensive education, with implications for subsequent careers and earnings (Anaya et al., 2022; Joensen and Nielsen, 2016). Research examines how gender differences in mathematics performance, particularly among high-achieving students, may contribute to these educational and occupational disparities (Ellison and Swanson, 2023; Niederle and Vesterlund, 2010). Understanding the conditions that support women's progress in mathematics therefore has implications beyond the classroom.

A substantial literature examines how educational environments shape women's mathematics achievement and continued participation. School-based studies investigate the roles of confidence, teachers' gender stereotypes, willingness to compete, and responses to setbacks (Bharadwaj et al., 2016; Buser and Yuan, 2019; Buser et al., 2014; Carlana, 2019; Ellison and Swanson, 2023). At the university level, research links classroom environment, cooperation, and belongings to mathematics learning experiences and achievements (Ellis et al., 2016; Ernest et al., 2019; Good et al., 2012; MacArthur and Dobie, 2023). This paper examines classroom gender composition as one dimension of that environment. I investigate whether greater exposure to female classmates improves women's completion of introductory mathematics and whether these effects extend to broader first-term and subsequent academic progress, particularly among students who enter university intending to pursue STEM.

Introductory mathematics courses in university serve as prerequisites for subsequent coursework and fulfill degree requirements across quantitatively oriented programs. Students' preparation and initial placement shape the courses available to them, while successful completion enables further progress through their programs. The first university term is also a period in which students establish new academic relationships and gather information about academic progress (Griffith and Main, 2019). Gender composition of the peer group may influence this transition through opportunities for academic support and participation, as well as students' assessments of their own capabilities (Borbely et al., 2023; Eisenkopf et al., 2015; Eren, 2017; Hill, 2015; Kessels and Hannover, 2008; Lavy and Schlosser, 2011; Lu and Anderson, 2015; Luo and Yang, 2023; Mouganie and Wang, 2020; Sullivan, 2009)[1]. The gender-specific effects might especially salient in an environment related to

[1]Academic support can come from improvement in teacher-student relationship (Lavy and Schlosser, 2011), cooperative learning (Lu and Anderson, 2015), and daily information transmissions (Luo and Yang, 2023). Participation involves two sides of effects: Borbely et al. (2023) and Eren (2017) suggest more female classmates improve classroom

gender expectations (Bordalo et al., 2019; Coffman, 2014; Exley and Kessler, 2022).

The analysis uses administrative records for freshmen entering a selective U.S. public university between 2016 and 2023 and focuses on introductory mathematics courses taken during their first quarter. The identification strategy exploits variation in female peer share across classes taking the same course with the same instructor in the same term. Course–term–instructor fixed effects are interacted with student gender to account for shared classroom environment and the differential association with women's and men's outcomes. Identification assumes that, conditional on observed student and classroom characteristics, the remaining variation in female peer share is unrelated to unobserved determinants of academic performances.

I assess this assumption by examining whether female peer share is associated with classroom demographic composition, mean high-school GPA, and meeting schedules. These checks reveal no consistent pattern linking female share to the observed classroom characteristics. The baseline estimates remain largely unchanged when these controls are excluded or their coefficients are allowed to differ by student gender, suggesting that observed classroom characteristics do not account for the main findings. To assess sensitivity to pandemic-related changes in enrollment and learning, I restrict the sample to pre-pandemic entry cohorts and obtain estimates comparable to those for the full sample.

A further concern is that students may select classes based on preferences or constraints related to both classroom gender composition and their academic outcomes. To investigate this potential sorting, I conduct three tests. First, I compare the observed distribution of classroom gender composition with distributions generated by simulated random assignments (Bietenbeck, 2020; Dewan et al., 2024; Getik and Meier, 2022). Second, I test whether predetermined student characteristics are correlated with the leave-one-out female share in a student's class (Guryan et al., 2009; Zhang, 2025). Third, I use permutation tests to compare the observed balance-test coefficients with those obtained under simulated random assignments (Caeyers and Fafchamps, 2026). Together, these exercises provide limited evidence of systematic sorting based on observed characteristics and support the identifying assumption.

The baseline estimates indicate that greater female peer exposure improves women's early academic progress. A one-standard-deviation (13.7 percentage point) increase in female peer share raises the probability of completing introductory mathematics by 4.3 percentage points and the probability of earning C- or better by 4.6 percentage points. Both estimates are statistically signif-

participation and higher attendance; while Lavy and Schlosser (2011) and Hill (2015) shows that more male display disruptive behaviors and more opposite-gender relationship undermine students' achievement, respectively. Kessels and Hannover (2008), Sullivan (2009), Eisenkopf et al. (2015), and Mouganie and Wang (2020) associate the peer composition to gender-related self-knowledge, academic self-assessment, and affirmation.

icant at the 5 percent level. For a 10-percentage-point increase in female peer share, these effects correspond to increases of 3.1 and 3.4 percentage points, respectively, or 3.8 and 4.2 percent relative to the corresponding outcome means. Female peer exposure also improves women's entry-term outcomes. Men experience improvements in broader first-quarter completion and grade-threshold attainment, although their mathematics-specific estimates are less precise. Most comparisons do not reject equality of the effects across genders. The findings remain robust to alternative control specifications, accounting for gender composition in other courses and parallel mathematics classes. The benefits for women are concentrated among students who declared a STEM major at entry. Longer-term estimates also provide suggestive evidence of increased STEM degree attainment among women and men who entered with a declared STEM major.

Five potential mechanisms may link female peer exposure to improvements in women's academic outcomes (Oosterbeek and van Ewijk, 2014). First, instructors may respond to classroom gender composition through their teaching practices and interactions with students (Carlana, 2019; Carrell et al., 2010; Dee, 2007; Griffith, 2014; Hoffmann and Oreopoulos, 2009; Lavy and Schlosser, 2011; Riegle-Crumb and Humphries, 2012; Spilt et al., 2012). Second, students may benefit from learning spillovers associated with peer academic ability (Booij et al., 2017; Coveney and Oosterveen, 2021; Feld and Zölitz, 2017; Kimbrough et al., 2022). Third, greater female representation may strengthen women's confidence and sense of belonging in settings shaped by stereotypes about mathematical and STEM abilities (Goetz et al., 2013; Good et al., 2012; Jouini et al., 2018; Mann and DiPrete, 2016; Niederle and Vesterlund, 2010). Fourth, female peers may facilitate participation, cooperation, and supportive academic relationships, channels whose relevance to university mathematics is highlighted by research on classroom interactions (MacArthur and Dobie, 2023; Reinholz et al., 2022). Fifth, lower male representation may reduce disruptive behavior and improve the learning environment (Lavy and Schlosser, 2011). The research design and empirical patterns help assess these potential explanations.

The analysis compares classes within course–term–instructor groups and allows these fixed effects to differ by student gender. This specification accounts for instructor characteristics and instructional conditions shared across classes, including their differential associations with women's and men's outcomes. It therefore limits the role of differences across instructors' race and gender in explaining the findings. The administrative records also contain freshmen's high-school GPA, a predetermined measure of academic preparation and a proxy for ability at university entry. The estimates remain stable after controlling for classmates' leave-one-out mean high-school GPA, suggesting that differences in observed peer ability do not account for the benefits of greater female

peer exposure.

Although the data do not directly measure psychological and behavioral responses, this research provide empirical patterns informing the interpretation of these channels. The findings remain robust when the sample is restricted to pre-pandemic cohorts, and excluding the 2020 cohort, whose first-term courses were delivered online, also leaves the estimates broadly unchanged. These patterns limit explanations that attribute the findings primarily to pandemic-related changes in interactions among students and between students and instructors. On the other hand, comparable positive effects for women's academic performance in mathematics do not emerge in basic writing and introductory English, or other entry-term courses. Within introductory mathematics, the benefits for women are concentrated among freshmen who declared a STEM major at entry. These patterns are consistent with the possibility that greater female representation eases psychological strain and strengthens academic confidence in a setting where gender stereotypes may be salient and mathematics performance is consequential for students' intended degree paths (Niederle and Vesterlund, 2010; Steele, 1997). This interpretation complements evidence linking female peer exposure to more supportive classroom interactions and lower perceived academic interference from worry and anxiety (Griselda et al., 2026).

This paper makes several contributions to the literature. First, it adds to research on peer gender composition in higher education by examining foundational mathematics courses taken by first-term freshmen from multiple majors at a U.S. university. Existing studies often focus on particular faculties or degree programs, including undergraduate economics and business (Booth et al., 2018; Oosterbeek and van Ewijk, 2014; Shan, 2026; Zölitz and Feld, 2021), arts and social sciences (Zhang, 2025), engineer (Griffith and Main, 2019), doctoral STEM programs (Bostwick and Weinberg, 2022), and MBA programs (Hampole et al., 2026). Findings for women and men vary across these settings.[2] My setting brings together students with different intended academic trajectories in the same classroom, allowing me to examine responses to female peer exposure among students entering with declared STEM and non-STEM majors.

The paper also provides classroom-level evidence on gender peer effects in U.S. undergraduate education through comparisons across classes taught by the same instructor for the same course in the same term. Previous literature exploit variations within colleges or programs (Aguirre et al., 2026; Bostwick and Weinberg, 2022; Hill, 2017), variations across sections or tutorial groups within courses (Griffith and Main, 2019; Griselda et al., 2026; Hampole et al., 2026; Zhang, 2025; Zölitz and Feld, 2021), or random assignment across workgroups or classes (Booth et al., 2018; Oosterbeek

[2]Table A.1 summarizes these studies and their findings.

and van Ewijk, 2014; Shan, 2026). My analysis compares students taking the same introductory mathematics course with the same instructor in their first university term. Gender-specific course–term–instructor fixed effects account for shared instructional conditions and allow their associations with academic outcomes to differ between women and men. This feature is relevant given evidence that the gender or race pairing of instructors and students can influence academic achievement and educational choices (Bettinger and Long, 2005; Carrell et al., 2010; Griffith, 2014; Hoffmann and Oreopoulos, 2009). Balance tests and comparisons with simulated random assignments provide evidence consistent with the absence of systematic sorting across classes within these groups. By studying variation arising under an existing course-registration system, the paper provides evidence on gender peer effects in ordinary university classrooms, highlighting the relevance of classroom composition for course-allocation practices and policies supporting students' academic progress.

More broadly, this paper contributes to understanding how university peer environments influence academic outcomes. Existing research examines peers in residential settingvcbs, particularly roommates (Barwick et al., 2026; Booij et al., 2017; Carrell et al., 2008; Chen and Hu, 2024; Feld and Zölitz, 2017; Kremer and Levy, 2008; Sacerdote, 2001; Stinebrickner and Stinebrickner, 2006, 2008; Zimmerman, 2003), and in classrooms, where students share instruction and academic tasks (Booij et al., 2017; Feld and Zölitz, 2017; Feld and Zölitz, 2022; De Giorgi and Pellizzari, 2014). Previous research using peer composition in the first-term university focuses on a specific field of study (Anelli et al., 2023; Fischer, 2017; Griffith and Main, 2019; Ost, 2010).

I extend the classroom literature by examining gender composition in introductory mathematics, a setting closely related to that of Anelli et al. (2023), who study how foreign classmates affect domestic students' subsequent STEM attainment. In this paper, the estimates remain stable after controlling for female peer share in other courses. Separate analyses of basic writing and introductory English courses, and of the remaining entry-term courses outside these subject, do not reveal comparable positive effects for women. These findings highlight the importance of examining peer effects within specific fields of study; particularly, this paper suggests that classroom gender composition may be consequential in settings shaped by gendered expectations, with implications for gender disparities in academic progress across STEM and non-STEM fields in higher education.

Finally, this paper contributes to understanding how a particular educational environments shape women's academic performance and persistence in mathematics and science. To explain the gender differences in mathematics performances, many research points to cultural environments (Dossi et al., 2021; Lippmann and Senik, 2018; Nollenberger et al., 2016; Pope and Sydnor, 2010; Spencer et al., 1999).[3] Within the school setting, curriculum policies and gender stereotypes of

[3]Family preferences for boys (Dossi et al., 2021), culturally transmitted gender norms (Nollenberger et al., 2016;

teachers can also influence mathematics-related achievement and educational choices (Carlana, 2019; Joensen and Nielsen, 2016; Riegle-Crumb and Humphries, 2012).

Studies of mathematics learning environment bring these questions to university. Research documents the low visibility of women's participation and difficulties in women's collaboration with peers in mathematics classes (Ernest et al., 2019; MacArthur and Dobie, 2023; Reinholz et al., 2022). And others associate belonging and confidence with women's mathematics achievement and willingness to continue studying mathematics (Ellis et al., 2016; Good et al., 2012). As university is an environment where major choice has long-lasting influence on subsequent careers and earnings (Altonji et al., 2016; Kirkebøen et al., 2016; Stinebrickner and Stinebrickner, 2014), and research also connects mathematics performances to selection on mathematics- and science-intensive academic tracks and future earnings (Altonji et al., 2012; Anaya et al., 2022; Buser et al., 2014; Card and Payne, 2021; Delaney and Devereux, 2019, 2020, 2021; Dougherty et al., 2017; Joensen and Nielsen, 2009; Niederle and Vesterlund, 2010). Building on evidence that female role models and peers can strengthen women's persistence in university STEM fields (Dasgupta et al., 2015; Dennehy and Dasgupta, 2017; Fiala et al., 2022; Fischer, 2017; Murphy et al., 2007; Stout et al., 2011), I provide evidence from first-term introductory mathematics courses that greater female peer exposure improves women's course completion and broader first-term academic progress, particularly among students who declared a STEM major at entry. These findings highlight the relevance of classroom peers in a setting where gender stereotypes may be salient and mathematics performance is consequential for students' intended degree paths.

The remainder of this paper is organized as follows. Section 2 introduces institutional background, data, and sample. Section 3 outlines the empirical strategy. Section 4 presents the main results, and Section 5 explores mechanisms. Section 6 concludes.

2 Data

2.1 Institutional Background

I use administrative student records from a selective public university in the United States. The university operates on a quarter system, with regular instruction in the fall, winter, and spring. The records include students' course registration histories, which allow me to reconstruct class rosters. Each class is identified by its subject, course title, instructor, and academic term. The data also provide students' gender, race and ethnicity, high school grade point average (GPA), and citizenship

Pope and Sydnor, 2010), broader social expectations concerning women's roles (Lippmann and Senik, 2018; Spencer et al., 1999) help explain differences in mathematics performance.

status. Academic records include term-specific declared majors, course grades, cumulative GPA, and, where observed, graduation dates and degrees awarded.

The university serves a large, racially and ethnically diverse undergraduate population and offers bachelor's degrees in more than 100 majors. Students may enter without a declared major but must select one by the end of their sophomore year. Changes of major require approval from advisors in both the originating and receiving programs. Incoming freshmen receive academic advising and register for their initial fall-quarter courses before instruction begins, generally during orientation.

Mathematics occupies a central position in the undergraduate curriculum at this university. Although the university does not prescribe a single mathematics course for every undergraduate, mathematics is included in the college breadth or major requirements of most degree programs. STEM programs—including engineering, physical sciences, biological sciences, computer science, and mathematics—typically require structured mathematics sequences as prerequisites for later coursework. Mathematics is also required in many non-STEM or social-science programs, including economics, business, psychology, and public policy, where students need quantitative preparation for calculus, statistics, research methods, or data analysis. Introductory mathematics courses therefore serve a broad population of students and frequently operate as gateways to subsequent courses and degree progress.

Introductory mathematics courses form an early component of degree requirements across engineering, computing, life sciences, economics, and business administration. Students' initial mathematics enrollment reflects both their academic preparation and their degree requirements. In my sample, over 98% who graduated with a STEM degree and over half who graduated with a non-STEM degree have once taken the introductory courses. Preparatory courses establish the foundation for calculus, while calculus sequences provide the quantitative training required for subsequent coursework. The specific course eligibility depends on Mathematics Advisory Examination placement, Advanced Placement credit, and previously completed coursework. College restrictions and major requirements further determine the courses available to each student. Students who need additional preparation begin with precalculus or a science-oriented preparatory sequence, whereas others enter standard calculus, life-sciences calculus, or business calculus. Qualifying prior credit also allows some freshmen to begin with later courses in a calculus sequence. Enrollment decisions therefore have implications beyond the first term: postponing a required mathematics course may delay access to later courses and disrupt the planned sequence of study.

A mathematics course may offer several lecture sections in the same term. Throughout this paper, students in a class refers to students of a lecture section and its associated discussion

sections. The lecture instructor is responsible for the syllabus and final course grades, while teaching assistants lead smaller discussion sections that supplement the lectures. The number of discussion sections varies with class size. A lecture class may have up to four teaching assistants, each leading one or two discussion sections of approximately 10–30 students. Students sharing a lecture class therefore receive instruction from the same instructor but may attend different discussion sections. Lecture schedules differ across classes. Some classes meet twice per week for 80 minutes per meeting, whereas others meet three times per week for 50 minutes. Lecture start times range from 8:00 a.m. to 6:40 p.m. Students must accommodate both lecture and discussion meetings within their broader course schedules.

The department assigns both lecture instructors and teaching assistants, but these assignments occur at different stages. Lecture instructors are generally designated earlier, and their assignments are typically posted before those of teaching assistants. Teaching assistants are usually assigned shortly before the quarter begins, usually after incoming freshmen have selected their initial schedules. Students consequently have limited information about their assigned teaching assistants when they first register, reducing the scope for initial section selection based on TA identity.

Students may add or drop courses and change sections during the first three weeks of each quarter. According to academic advising staff, changes to introductory mathematics enrollment are uncommon because students must coordinate their schedules and maintain progress toward subsequent requirements. The costs of adjustment differ by the type of change: postponing a required mathematics course may delay access to advanced coursework, while switching sections of the same course may require rearranging other classes. These constraints can discourage enrollment changes.

2.2 Sample and Description Statistics

The analysis focuses on first-term freshmen enrolled in introductory mathematics courses during the 2016–2023 entry cohorts. Focusing on the introductory mathematics in the first term of freshmen limits the scope for section selection based on prior experience with university instructors or established campus peer networks. Eligible courses are those that students can enter without prior coursework at the university, subject to placement requirements or qualifying prior credit. The sample spans preparatory mathematics, standard calculus, life-sciences calculus, and business calculus. Table A.3 summarizes enrollment and class composition across course categories. The 243 classes contain 17,477 freshmen, who account for approximately 72 percent of total enrollment. The average female share among freshmen is 46 percent, ranging from 29.7 percent in Standard

Calculus III to 57.3 percent in Life-sciences Calculus I. These differences reflect the distinct student populations served by different mathematics pathways.

Table 1 describes the organization and composition of courses, instructor groups, and classes. A class is defined as a credit-provided lecture of a course offered in a given term. The sample contains 72 course–term combinations, 186 course–term–instructor groups, and 235 classes. Each course–term combination enrolls an average of 336 students across 3.38 classes taught by 2.58 instructors. Within a course and term, an instructor teaches an average of 1.31 classes comprising around 131 students. The average class contains approximately 101 students, including 72 freshmen. The empirical analysis compares female peer exposure across classes of the same course taught by the same instructor in the same term, holding these three dimensions constant. Across all classes, the female share averages 46.01 percent, with a standard deviation of 15.41 percentage points.

Identification relies on course–term–instructor groups containing multiple classes with different gender compositions. Groups containing only one class provide no between-class variation in female peer exposure and therefore do not directly identify the peer effect.[4] I nevertheless retain these observations because they can contribute to estimating coefficients on individual-level controls. The sample supplying identifying variation is 110 of the 243 classes, or approximately 45 percent, contributing to identification.[5] Figure 1 showing residualized leave-one-out female shares illustrates the variation remaining after removing course–term–instructor fixed effects. This within-group variation accounts for approximately 31 percent of the total variance in female peer exposure. Figure B.1 presents the corresponding distribution for the subsample supplying identifying variation, in which approximately 47 percent of the total variance remains after removing these fixed effects.

Table 2 reports student-level summary statistics for the 8,916 male and 8,561 female freshmen in the analysis sample. Panel A presents predetermined characteristics, including race and ethnicity, international student status, high school GPA, and STEM major declaration at entry. Female students have a slightly higher average high school GPA than male students, at 3.82 compared with 3.76, but are less likely to enter with a declared STEM major. Because the available follow-up period differs across cohorts, the main analysis examines academic outcomes during the first term, summarized in Panel B. Women complete a slightly larger share of their first-term courses and are more likely to earn grades of C- or better in every course, at 52 percent compared with 48 percent

[4] Anelli et al. (2023) primarily exploit changes in peer composition within course–instructor pairs over time. They also consider comparisons within course–term–instructor groups, but few instructors in their sample teach multiple sections of the same course in a given term, limiting statistical power. The present study uses this more restrictive comparison as its primary source of identifying variation.

[5] Table A.4 summarizes the composition of courses and classes contributing to identification in the main specification. These classes represent a larger share of the sample than in the corresponding specification of Anelli et al. (2023), where approximately 7 percent provide identifying variation.

for men.[6] Except for that, women and men see little differences in other outcomes including average first-term GPA.

To examine longer-term academic outcomes, I restrict the sample to the 2016–-2020 entry cohorts, for whom a longer follow-up period is available. Table A.5 compares their academic outcomes with those of the full 2016–2023 sample. Panel A shows that short-term outcomes are broadly similar across the two samples within each gender: mathematics completion rates, the probability of earning a grade above C-, and first-term course completion differ by no more than one percentage point. However, average first-term GPA is somewhat lower among the earlier cohorts, at 3.16 for female and 3.15 for male students, compared with 3.20 in the full sample for both genders. In the following results, I display that the female share effects on short-term academic outcomes are comparable across 2016-2020 and 2016-2023 cohorts. Panel B summarizes longer-term outcomes for the 2016–2020 cohorts, including time to declare a major, dropout, and graduation within four academic years. Female students have a lower dropout rate than male students, 19 versus 24 percent, and a higher four-year graduation rate, 78 versus 72 percent. However, women are less likely to graduate with a STEM degree within four years, at 43 percent compared with 49 percent for men.

3 Empirical Strategy

I estimate the effect of exposure to female classmates on first-term freshmen's course performance and subsequent academic outcomes. The empirical strategy compares students enrolled in different classes of the same mathematics course taught by the same instructor in the same quarter. By holding the course, instructor, and term constant, this comparison removes differences in peer composition associated with course selection, instructor choice, and common term-specific conditions. Identification relies on the assumption that the remaining variation in female peer exposure is unrelated to unobserved determinants of students' outcomes, conditional on the included controls.

I estimate the impact of exposure to female students share using a linear model:

$$\begin{aligned} Y_{ikct} = {} & \alpha_1 \overline{\text{FemaleShare}}_{-i,kct} \times \text{Male}_i + \alpha_2 \overline{\text{FemaleShare}}_{-i,kct} \times \text{Female}_i \\ & + \sum_{j=0,1} \theta_{g(k)ct}\, I(\text{Female}_i = j) + X_{ikct}\eta' + Z_{-i,kct}\gamma' + S_{kct}\delta' + \varepsilon_{ikct}. \end{aligned} \tag{1}$$

where Y_{ikct} represents an outcome for freshmen i who attended a class c in an introductory math

[6] A grade of C- or better is commonly required in prerequisite courses to enroll in more advanced coursework. I examine indicators for earning grades of A- or better and B- or better to capture both meeting this threshold and attaining high academic performance.

course c in term t. The main outcomes of interests include the indicator for mathematics course completion, and the mathematics course with grade higher than C-, the course of completion at the entry term, and the share of courses with grade higher C-, since grades higher than C- are the requirements for the subsequent coursework. I emphasize course completion and the share of courses with grades of C- or better because the analysis focuses on persistence and progression through first-term coursework. These measures retain withdrawn courses in the denominator and therefore capture a dimension of academic progress that GPA does not directly reflect. I nevertheless report GPA as the supplementary results.

Female peer exposure is defined as the proportion of female classmates, excluding the student i from the calculation. I standardize this leave-one-out share and interact it with female and male indicators allowing the effects to differ by gender. The primary coefficients α_1 and α_2 therefore measure the effects of a one-standard-deviation increase in female peer exposure on male and female students, respectively.

The model includes course–term–instructor fixed effects, $\theta_{g(k)ct}$, interacted with student gender. These fixed effects allow average outcomes for men and women to differ across course–term–instructor groups, accounting, for example, for differences in how students of each gender respond to a particular course or instructor. Identification therefore comes from differences in female peer exposure across classes within these groups, separately for male and female students.

Individual controls, X_{ikct}, include age, indicators for White, minority, and international students, and high school GPA. Class-level controls, $Z_{-i,kct}$, include the shares of White, Asian, Black, Latino, and international classmates and classmates' mean high-school GPA, all calculated excluding student i.These controls account for observed differences in peer composition that may accompany differences in female representation. I also control for the class-schedule indicators, S_{kct}. Since instructional schedules and time of day can affect academic performance (Cotti et al., 2018; Gaggero and Tommasi, 2023; Williams and Shapiro, 2018), and temporal flexibility can be particularly important for women with childcare and household responsibilities (Goldin, 2014), classroom gender composition may therefore be correlated with scheduling characteristics that also influence academic outcomes. To account for these potential relationships, I include a categorical variable combining twice-weekly meeting status with three class start-time categories: early morning (8:00–9:59 a.m.), late afternoon (4:00–6:59 p.m.), and all other times. This variable captures combinations of meeting frequency and time of day that may influence students' scheduling choices, particularly when early-morning or late-afternoon classes conflict with household or childcare responsibilities. Standard errors are clustered at the course–term–instructor level.

3.1 Identification Challenge

Identification assumes that, among students taking the same mathematics course with the same instructor in the same term, differences in female peer share are unrelated to unobserved factors affecting academic outcomes, conditional on observed student and classroom characteristics. These comparisons account for shared instructional conditions while allowing their effects to differ between women and men. The remainder of this section explains the identification challenges and how the analysis addresses them.

Reflection. The conventional reflection problem arises when an individual's outcome depends on peers' contemporaneous outcomes, which may themselves respond to the individual. The explanatory variable here is classmates' gender composition, a predetermined characteristic, rather than their academic performance. The model therefore avoids including a simultaneously determined peer outcome as a regressor. The estimated coefficients capture the overall effect of female peer exposure, potentially operating through changes in peer behavior, classroom interactions, or other channels.

Common shocks. Course–term–instructor fixed effects absorb conditions shared by classes taught by the same instructor in a given course and term. Nevertheless, class-specific factors could confound the estimates if they are correlated with both female representation and student outcomes. For example, classes within the same group may differ in the composition of their students.

To assess observable differences across classes, I aggregate the data to the classroom level and regress the female share on classroom demographics, mean high-school GPA, class size, and scheduling indicators. Table 3 displays the results. As discussed in the description of Equation 1, gender differences in scheduling preferences, potentially reflecting household or childcare responsibilities, may link classroom gender composition to scheduling characteristics that also affect academic performance. I therefore examine interactions between an indicator for classes meeting twice per week and indicators for early-morning and late-afternoon classes, reported in Columns (8) and (9). All specifications include course–term–instructor fixed effects. None of the coefficients in the separate regressions in Columns (1)–(9) is statistically significant. Column (10) includes all characteristics jointly. Mean age is negatively associated with the female share at the 10 percent significance level, but the joint test fails to reject that all included coefficients are zero, with a p-value of 0.208. Overall, these results provide limited evidence of systematic sorting across classes based on observed characteristics, although they do not rule out selection based on unobserved scheduling preferences.

The COVID-19 pandemic may have affected both course enrollment and academic performance. To assess whether the baseline findings are sensitive to these disruptions, I reestimate the model

using only the 2016–2019 entry cohorts. These students completed their first-term courses by December 2019, so their short-term outcomes were measured before the pandemic. The estimated effects of female peer exposure are similar to those obtained from the full sample, indicating that the short-term findings are not driven by pandemic-period observations. Section 4 presents these results in detail.

Selection. Students may select classes according to their preferences, academic schedules, or academic preparation, potentially generating a relationship between female peer exposure and individual characteristics even within the same course–term–instructor group. I assess this concern by comparing observed classroom gender composition with a random-allocation benchmark and examining whether female peer exposure is associated with predetermined student characteristics.

To start with, I provide descriptive evidence examining whether the observed distribution of classroom gender composition is consistent with random allocation (Bietenbeck, 2020; Dewan et al., 2024; Getik and Meier, 2022). Within each course–term–instructor pool, I randomly reassign students across classes while preserving observed class sizes and the total number of female students in the pool. For each of 500 simulated allocations, I regress classroom female shares on course–term–instructor fixed effects and retain the residuals. I apply the same procedure to the observed allocation.

Figure 2 compares the distribution of residual female shares under the observed allocation, represented by the solid line, with the pooled distribution from the simulations, represented by the dashed line. The distributions closely resemble one another, and the reported distributional comparison yields a p-value of 0.58, failing to reject the null that they are identical. The observed dispersion in classroom gender composition is therefore compatible with the random-allocation benchmark. However, similarity in these distributions does not establish that students with different characteristics are equally likely to enter classes with higher female shares. I examine this possibility using student-level balance regressions and evaluate the resulting coefficients against their distributions under simulated random assignment.

The balance regressions relate each predetermined student characteristic to the leave-one-out female share in the student's class. Because students cannot be their own peers, excluding each student from the calculation of peer composition can generate mechanical relationships even under random assignment. Following Guryan et al. (2009), I include the leave-one-out female share in the assignment pool, defined as all students taking the same course with the same instructor in the same term.

A specification with a single coefficient on this correction term imposes the same adjustment

across pools, regardless of how many classes they contain. This restriction is potentially consequential in my setting, where the organization of classes differs across courses and instructors. Dividing a pool into two classes creates a different allocation structure from dividing it into several classes: the number of alternative peer groups and the proportion of potential peers encountered in a particular class can differ. Following Zhang (2025), I therefore interact the pool-level correction term with indicators for the number of classes in the pool. This specification allows the adjustment to vary with the structure of class offerings, rather than requiring a common coefficient across all pools or restricting the sample to pools with the same number of classes.

This flexibility addresses heterogeneity in the number of classes, but it does not necessarily resolve the separate identification concern discussed by Caeyers and Fafchamps (2026). Their concern arises when pool sizes—the total numbers of students available for assignment—vary little. I therefore supplement the regression-based diagnostics with permutation inference that preserves the observed allocation structure.

I examine the balance check using the following regression model:

$$X_{ikct} = \beta_1 \overline{\text{FemaleShare}}_{-i,kct} + \sum_{n=1}^{N} \beta_n \overline{\text{FemaleShare}}_{-i,g(k)ct} D_n + \theta_{g(k)ct} + \epsilon_{ikct}. \tag{2}$$

X_{ikct} represents measures of individual pre-determined characteristics of student i in the class k of the math course c and in term t. $\overline{\text{FemaleShare}}_{-i,g(k)ct}$ is the correction term proposed by Guryan et al. (2009) address the concerns of the mechanically correlation between student's characteristics and the leave-one-out female share in each class. As discussed above, I use the Zhang (2025)'s method controlling for the the number of classes, D_n, present within each term-course-instructor pool, n. The number of classes within a pool is distinct from pool size, which refers to the total number of students in that pool. I therefore interpret these regressions as balance diagnostics rather than definitive tests of random assignment. Similar to the baseline regression, I also control for the fixed effects at the term-course-instructor level, $\theta_{g(k)ct}$, and the standard errors are clustered at the term-course-instructor level.

Table 4 reports the balance results, with each column examining a different predetermined characteristic, including gender, race and ethnicity, international student status, age, and high school GPA. None of the coefficients on classroom female share is statistically distinguishable from zero at conventional levels. These estimates provide no detectable evidence that students' observed characteristics are systematically associated with female peer exposure within course–term–instructor groups.

Following the within-pool randomization approach of Caeyers and Fafchamps (2026), I assess

whether the observed association between student characteristics and female peer share is consistent with random classroom assignment. We generate 1,000 counterfactual allocations by randomly assigning students to the original classroom slots within admission-cohort × course × instructor pools, preserving classroom sizes and pool composition. For each allocation, I reconstruct leave-one-out female peer shares using the original peer roster and reestimate the same balance regression on the analysis sample. Table 3 also reports the resulting two-sided randomization p-values.[7] These tests do not reject the random-assignment benchmark for the characteristics examined: the observed coefficients are not unusually extreme relative to those generated by reassignment within pools.

To assess these findings using randomization inference, I reestimate the balance regressions for each simulated allocation and compare the observed coefficient on classroom female share with its simulated distribution. This procedure follows the permutation approach discussed by Caeyers and Fafchamps (2026), allowing the reference distribution to reflect the mechanical relationships generated by the observed pool and class sizes. Table 3 also reports the resulting two-sided randomization p-values. These tests do not reject the random-assignment benchmark for the characteristics examined: the observed coefficients are not unusually extreme relative to those generated by reassignment within pools.

4 Results

4.1 Baseline Results

Table 5 reports the estimated effects of female peer exposure on introductory mathematics performance and entry-term academic outcomes. All specifications include individual and classroom controls and course–term–instructor fixed effects interacted with student gender. The leave-one-out female peer share is standardized using the full-sample standard deviation of 0.136. Each coefficient therefore measures the effect of a 13.6-percentage-point increase in female peer share.

Greater female peer exposure improves women's mathematics performance. A one-standard-deviation increase in female peer share raises the probability of completing introductory mathematics by 4.3 percentage points and the probability of earning C- or better by 4.6 percentage points. Both estimates are statistically significant at the 5 percent level. For a 10-percentage-point

[7] Following Caeyers and Fafchamps (2024), I implement a two-sided equal-tail test: the p-value equals twice the smaller of the lower- and upper-tail probabilities of the observed coefficient, capped at one. Both probabilities include ties and are calculated as the corresponding simulation count plus one, divided by 1,001, applying the finite-simulation correction discussed by Phipson and Smyth (2010). This procedure evaluates the observed coefficient relative to its random-assignment distribution without assuming that the distribution is centered at zero

increase in female peer share, these effects correspond to increases of 3.1 and 3.4 percentage points, respectively, or 3.8 and 4.2 percent relative to the corresponding outcome means. Female peer exposure also improves women's entry-term outcomes.

The estimates for men are positive but smaller. A 10-percentage-point increase in female peer share raises the estimated probability of mathematics completion by 1.4 percentage points, and the probability of earning C- or better by 2.4 percentage points. Despite the larger point estimates for women, the formal tests do not reject equality of women's and men's effects for any of the four outcomes.

Table A.6 examines sensitivity to the inclusion and specification of controls. For each outcome, I compare estimates without controls, with common control coefficients, and with control coefficients that differ by gender. All specifications retain course–term–instructor fixed effects interacted with student gender. Women's estimated effects remain positive and statistically significant throughout, increasing modestly when control coefficients are allowed to differ by gender. Men's estimates remain positive, although their magnitude and statistical significance vary across specifications. Overall, the baseline findings persist after accounting for observed characteristics and allowing their associations with outcomes to differ by gender. Tests do not reject equality of women's and men's effects in any specification.

Table A.7 shows that greater female peer exposure in introductory mathematics improves academic progress both in mathematics and across the entry term. Within introductory mathematics, the benefits extend beyond completion to stronger passing grades, significantly increasing attainment of B- or better for both women and men. Greater exposure also increases the share of first-term courses completed and the share with grades of C- or better for both genders. However, the estimates for entry-term GPA are not statistically significant, providing no clear evidence of an improvement in average grades. Taken together, the results demonstrate gains in course completion and passing-grade attainment that support progress through foundational mathematics and access to subsequent quantitative coursework.

The female-share tercile analysis in Table A.8 examines whether the relationship varies across the exposure distribution. Relative to the lowest tercile, women in the highest tercile have higher probability in mid-term academic performances of interests compared to the middle tercile ones, although the results are not precisely estimated. For men, the highest tercile is associated with statistically significant increases in mathematics completion and in earning C- or better. Thus, the largest estimated gains generally occur in classes in the highest tercile of female representation.

4.2 Robustness Checks

Table 6 examines whether baseline findings reflect gender composition elsewhere in students' academic environments. I separately control for the average female share in students' other courses and in other classes within the same mathematics course–term–instructor group, allowing for these additional exposure coefficients to differ by gender. The estimated effects of female share in the focal mathematics class remain similar for both female and male students. These results suggest that the baseline relationship is not simply capturing correlated gender composition in other courses or classes.

Tables A.9 examines whether the findings extend beyond introductory mathematics. Panel A considers freshmen enrolled in basic writing and introductory English courses during their first term.[8] Panel B examines the remaining entry-term courses, excluding basic writing, introductory English, and introductory mathematics. If only statistically significant estimates are considered, the sole finding is a negative effect of female peer share on women's course completion in basic writing and introductory English. No statistically significant effects emerge for either women or men in the remaining courses. Thus, the positive effects on women's academic progress in introductory mathematics do not extend uniformly to other entry-term coursework.

Table A.10 and Figure 3 assess the sensitivity of the results to cohort coverage. Panel A reports estimates for the full 2016–2023 sample, while Panel B restricts the analysis to the 2016–2019 entry cohorts. Students in the latter sample completed their first fall term by December 2019, so their short-term outcomes precede the COVID-19 pandemic. The effects on women are comparable in these two samples, although mathematics completion and C–or-better coefficients are less precisely estimated. For men, the entry-term share estimates also remain similar, although the mathematics completion estimate becomes slightly negative and remains statistically insignificant. The principal positive entry-term findings therefore persist in a sample whose first-term instruction was unaffected by the pandemic.

Figure 3 assesses sensitivity to cohort composition by excluding one admission cohort at a time. The estimates remain broadly stable, suggesting that no single cohort drives the findings. In particular, the results persist when I exclude the 2020 cohort, whose entry-term courses were taught online. Together with the pre-pandemic estimates, these results suggest that the main findings are not driven by pandemic-related changes in the instructional environment.

[8]The basic writing and introductory English courses help students fulfill the university's undergraduate writing requirements. The basic writing courses include three courses, while introductory English courses include two courses.

4.3 Heterogeneous Results

Table 7 examines heterogeneity by students' declared major at entry. The positive effects for women are concentrated among students entering with a STEM major, for whom greater female peer exposure significantly increases mathematics completion and attainment of C- or better. Among male STEM entrants, the estimates are also positive, although only the effect on C- attainment is statistically significant. Among non-STEM entrants, neither women's nor men's estimates are statistically significant. These patterns suggest that female peer exposure supports women's progress in foundational mathematics particularly when it is directly relevant to their intended STEM degree paths.

Table A.11 reports estimates by race. The Asian subsample provides evidence of a gender difference in the response to female peer exposure: Asian women experience significant improvements in mathematics completion and attainment of C- or better, whereas Asian men's estimates are small and statistically insignificant. The formal comparison confirms a significantly larger effect for women on mathematics completion and C- attainment. Among White students, the estimates are negative for both genders, with a statistically significant decline only in men's C- attainment. In the minority subsample, the estimates are positive for both genders, with significant gains in both outcomes among men. Neither the White nor the minority subsample shows statistically significant gender differences.

Table A.12 suggests that the completion benefits of female peer exposure differ by prior academic preparation: women's estimated gains are larger in the lower-GPA group, whereas men's are larger in the higher-GPA group. Significant completion effects emerge for lower-GPA women and higher-GPA men. Lower-GPA women also experience a significant improvement in attainment of C- or better. These patterns suggest that female peer exposure may support students with different levels of preparation depending on gender.

4.4 Long-run Academic Outcomes

I next examine whether the short-term gains extend to subsequent academic trajectories. The longer-term analysis uses the 2016–2020 entry cohorts. Panel C of Table A.10 shows that the short-term estimates for these cohorts are similar to those for the full sample, preserving the principal entry-term findings.

Table 8 reports effects on major declaration, dropout, and graduation within four academic years. For both women and men, the estimates point toward later major declaration, lower dropout, and higher graduation probabilities, with no statistically significant gender differences in these out-

comes. Only the increase in men's time to major declaration is statistically significant. The degree-specific estimates suggest that greater female peer exposure shifts women's degree attainment toward STEM relative to men: women exhibit a larger estimated increase in STEM degree attainment and a decline in non-STEM degree attainment, whereas men's estimates are positive for both. The gender comparisons suggest that women and men respond differently to female peer exposure in their degree outcomes, with women exhibiting larger estimated gains in STEM degree attainment than men.

Table A.13 examines longer-term outcomes by major declared at entry. Among non-STEM entrants, greater female peer exposure significantly increases men's time to major declaration, while the remaining estimates are statistically insignificant. Among STEM entrants, the estimates point toward lower dropout and higher overall and STEM degree attainment for both women and men. Women exhibit a larger estimated increase in STEM degree attainment and a decline in non-STEM degree attainment, although none of these effects or the corresponding gender differences is statistically significant. Overall, the patterns are consistent with greater female peer exposure supporting persistence in STEM among students who entered with a STEM major.

5 Mechanism and Discussion

Female peer exposure affecting students' academic outcomes channeling through several potential mechanisms (Oosterbeek and van Ewijk, 2014). The research design and empirical patterns help assess these explanations before turning to the most potential channels of academic interaction and classroom climate.

Instructor-student interaction. Instructors may respond to classroom gender composition through their teaching practices and interactions with students. Related research highlights teacher–student relationships as an important dimension of the learning environment, documenting connections between gender composition and these relationships, as well as differences in how teachers relate to boys and girls (Lavy and Schlosser, 2011; Spilt et al., 2012). Teachers' gender stereotypes may also influence their assessments of students' mathematical ability, with implications for girls' confidence, achievement, and subsequent educational choices (Carlana, 2019; Riegle-Crumb and Humphries, 2012). Research on instructor–student gender pairing further examines how instructor gender affects students' achievement and educational decisions (Carrell et al., 2010; Dee, 2007; Griffith, 2014; Hoffmann and Oreopoulos, 2009).

My analysis compares classes taking the same course with the same instructor in the same term. Students in these classes therefore share the instructor's gender, background, and other

characteristics, limiting explanations based on differences in instructor characteristics or gender stereotypes across instructors. Interacting course–term–instructor fixed effects with student gender further allows these shared instructional conditions to have different associations with women's and men's outcomes. This also accounts for systematic differences related to instructor–student gender pairing. However, the same instructor can still adapt teaching practices or interactions with students to the female share in each class. Such responses remain a possible channel through which female peer exposure affects academic outcomes.

Learning spillovers. Students may benefit from peer-to-peer learning through discussion, explanations, and shared problem-solving. Experimental evidence demonstrates that peer teaching can improve learning, while research in university classrooms identifies group interaction as an important channel through which peer academic ability affects achievement (Booij et al., 2017; Coveney and Oosterveen, 2021; Feld and Zölitz, 2017; Kimbrough et al., 2022). To assess whether differences in peer preparation explain the benefits of female peer exposure, I compare specifications with and without controls for classmates' leave-one-out mean high-school GPA, a proxy for academic ability at university entry. Table A.6 shows that the estimates remain broadly stable across specifications without controls, with common controls, and with controls whose coefficients vary by student gender. This stability suggests that differences in observed peer preparation do not account for women's improved outcomes under greater female peer exposure. Furthermore, the estimates by students' own high-school GPA in Table A.12 reveal no clear pattern of systematically larger benefits among students with higher or lower prior achievement. Together, these findings provide limited support for an explanation based on differences in measured academic preparation.

Classroom interactions. Classroom gender composition may affect academic outcomes by changing how students interact. Hill (2015) discusses attention and opposite-gender relationships as possible channels linking peer gender composition to achievement, while Lavy and Schlosser (2011) finds that greater female representation reduces classroom disruption and improves the learning environment. Although my administrative records do not directly measure students' academic networks or interactions inside and outside the classroom, variation in instructional conditions provides context for interpreting these channels.

This analysis has sample covering the 2016–2023 entry cohorts, including the 2020 cohort, whose first-quarter courses were delivered online. Existing research documents gender differences in responses to instructional format and points to opportunities for academic interaction and support as potential explanations. For instance, Shan et al. (2025) finds that greater exposure to online instruction reduces women's performance, particularly in mathematics-intensive courses, whereas

Figlio et al. (2013) finds stronger evidence favoring live instruction among men. Kofoed et al. (2024) also documents lower grades under online instruction, alongside difficulties concentrating and weaker connections with classmates and instructors. Nevertheless, productive peer interaction can occur remotely: Song et al. (2015) finds that greater female representation improves productivity in online university teams, with suggestive evidence that collaborative communication contributes to these gains.

In my analysis, the estimates remain broadly stable when the sample is restricted to pre-pandemic cohorts, as shown in Table A.10. The leave-one-cohort-out estimates in Figure 3 also remain similar when the 2020 cohort is excluded. These findings show that the estimated benefits of female peer exposure persist under conventional in-person instruction and are not driven by the inclusion of cohorts exposed to pandemic-related disruptions.

Cooperation and participation. Greater female representation may strengthen academic interactions that support learning. Small-group learning improves achievement and persistence in undergraduate STEM education (Springer et al., 1999), and evidence from university settings connects female representation to greater sharing of notes, assistance with course content, and collaborative communication (Song et al., 2015; Griselda et al., 2026). Evidence from MBA sections further identifies information sharing and perceived support from male peers as potential channels linking female representation to women's subsequent career advancement (Hampole et al., 2026).

Participation offers a complementary channel (Zhang, 2025). Research documents gender differences in classroom speaking and interruption patterns (Crawford and MacLeod, 1990; Lee and McCabe, 2021; Sandler et al., 1996), including the limited visibility of women's contributions in undergraduate mathematics and associations between participation and mathematical performance (Ernest et al., 2019; Reinholz et al., 2022). Experimental evidence shows that male-dominated settings can weaken women's belonging and willingness to participate, while female-majority engineering groups encourage their participation (Dasgupta et al., 2015; Murphy et al., 2007). Greater opportunities to ask questions, exchange explanations, and obtain assistance could support the improvements in mathematics completion and grade-threshold attainment documented in this study.

Classroom norms and psychological strain. Female peer exposure may also change how students experience achievement, status, and evaluation. This channel concerns the social pressures surrounding academic work, rather than changes in formal grading incentives. Griselda et al. (2026) provide direct university evidence that greater female representation reduces conformity to masculinity-related attitudes concerning winning, risk-taking, and status. These changes may reduce the perceived social costs of acknowledging difficulty and seeking assistance. Related evidence

shows that making a dependency-related gender stereotype salient reduces women's willingness to seek help (Wakefield et al., 2012), while conformity to particular masculine norms is associated with less favorable psychological help-seeking (Wong et al., 2017).

A more supportive classroom climate may also reduce psychological barriers to engagement. Griselda et al. (2026) find that greater female representation reduces the perceived academic interference of worry and anxiety. In engineering, female representation in work groups reduces anxiety among first-year women (Dasgupta et al., 2015), and female peer mentors protect belonging and confidence and improve retention (Dennehy and Dasgupta, 2017). These findings relate closely to evidence from university mathematics linking belonging and confidence to achievement and continued participation in calculus (Ellis et al., 2016; Good et al., 2012).

These processes may be particularly consequential during the transition to university, when students establish new relationships while completing foundational requirements. In my setting, the benefits for women are concentrated in introductory mathematics and among students who declared a STEM major at entry, while comparable positive effects do not emerge in other entry-term courses. These patterns are consistent with a supportive peer environment helping women sustain engagement in mathematics courses that matter for their intended degree paths. The positive estimates for men also accommodate improvements in classroom support and climate that extend beyond women.

6 Conclusion

This paper examines how classroom gender composition affects students' academic progress during the transition to university. Using administrative records for freshmen entering a selective U.S. public university between 2016 and 2023, I exploit variation in female peer share across introductory mathematics classes within the same course–term–instructor group. Greater female peer exposure improves women's mathematics performance and broader first-term academic progress. A 10-percentage-point increase in female peer share raises women's probability of completing introductory mathematics by approximately 3.1 percentage points and earning math grade of C- or better by approximately 3.4 percentage points. Men experience improvements in broader first-term outcomes, and most formal comparisons do not reject equality of the effects across genders.

The findings remain robust to alternative control specifications, accounting for gender composition in other courses and parallel mathematics classes, and excluding pandemic-entry cohorts. The benefits for women are concentrated among students who declared a STEM major at entry, while comparable positive effects do not emerge in basic writing, introductory English, or other

entry-term courses. Longer-term estimates provide suggestive evidence of increased STEM degree attainment among both women and men who entered with a declared STEM major, although these estimates are less precise. Together, the results highlight the relevance of the peer environment in foundational mathematics courses for students' progress along their intended educational paths.

These findings have implications for efforts to support women's participation and persistence in STEM. Increasing female representation may benefit classmates as well as the students directly affected by enrollment policies. Universities should therefore consider the peer environments students encounter in foundational courses alongside broader goals for female enrollment. The concentration of benefits among women already intending to pursue STEM further emphasizes the importance of supporting academic progress after students enter these fields.

A limitation of this research is to identify the specific classroom interactions and psychological changes underlying these effects. The administrative records establish patterns in academic outcomes but do not directly measure cooperation, participation, help-seeking, confidence, belonging, or anxiety. Combining academic records with classroom observations, peer-network information, and repeated student surveys would help determine whether greater female representation improves access to academic support, changes perceptions of competition, or reduces psychological strain. Directly testing these mechanisms would clarify how classroom gender composition influences learning and inform interventions that support students' progress in mathematics and STEM.

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Table 1: Summary Statistics at Different Aggregation Levels

	Course (1)	Course–Instructor (2)	Class (3)
Size	339.31	131.34	100.53
	(136.25)	(83.51)	(42.71)
Freshmen Size	243.42	94.23	72.12
	(150.41)	(86.15)	(47.22)
Female Share (%)	47.27	46.52	46.03
	(12.95)	(14.91)	(15.26)
White Share (%)	10.07	10.52	10.44
	(3.28)	(4.53)	(5.26)
Asian Share (%)	59.66	58.56	57.90
	(14.17)	(15.24)	(15.51)
Black Share (%)	3.68	3.67	3.68
	(2.12)	(2.72)	(2.99)
Latino Share (%)	24.84	25.15	26.00
	(12.57)	(13.15)	(13.28)
International Share (%)	3.21	3.41	3.35
	(3.15)	(4.05)	(4.27)
High School GPA	3.80	3.81	3.81
	(0.12)	(0.12)	(0.12)
# Instructors	2.58	1.00	1.00
	(1.18)	(0.00)	(0.00)
# Classes	3.38	1.31	
	(1.45)	(0.51)	
Counts	72	186	243

Notes: The sample consists of 17,477 freshmen enrolled in introductory mathematics courses during their first term across the 2016–2023 entry cohorts. Columns (1), (2), and (3) report means at the course, course–instructor, and class levels, respectively. The class is defined as the lecture sections within each course-instructor pairs. Standard deviations are in parentheses. The final row reports the number counted at each level of aggregation.

Table 2: Summary Statistics of Freshmen in Introductory Mathematics Courses

	Male (1)	Female (2)
Panel A. Background Characteristics		
White	0.12	0.099
	(0.32)	(0.30)
Asian	0.56	0.51
	(0.50)	(0.50)
Black	0.035	0.049
	(0.18)	(0.22)
Latino	0.27	0.32
	(0.44)	(0.47)
International	0.035	0.021
	(0.18)	(0.14)
High School GPA	3.76	3.82
	(0.30)	(0.28)
Declared STEM at Entry	0.77	0.71
	(0.42)	(0.45)
Panel B. Academic Outcome		
Completed Math Course	0.83	0.84
	(0.37)	(0.37)
Grade Higher than C-	0.80	0.80
	(0.40)	(0.40)
Share of Courses Completed in Entry Term	0.86	0.87
	(0.18)	(0.17)
Share of Courses with Grade Higher than C- in Entry Term	0.48	0.52
	(0.50)	(0.50)
Entry Term GPA	3.20	3.20
	(0.68)	(0.66)
Observations	8,916	8,561

Notes: The sample consists of 17,477 freshmen enrolled in introductory mathematics courses during their first term across the 2016–2023 entry cohorts. Columns (1) and (2) report class-level means for male and female students, respectively. Panel A summarizes students' demographic characteristics and educational backgrounds. Panel B presents the academic outcomes examined in this study. Standard deviations are reported in parentheses. Mathematics Courses are listed in Table A.3.

Table 3: Correlation between Class Gender Composition and Other Class Characteristics

	Dependent variable: Classroom female share									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
White Share	-0.0184									-0.859
	(0.240)									(0.575)
Asian Share		-0.153								-0.771
		(0.118)								(0.494)
Black Share			0.132							-0.308
			(0.460)							(0.517)
Latino Share				0.206						-0.538
				(0.228)						(0.454)
Mean Age					-0.293					-0.341*
					(0.237)					(0.182)
Mean High School GPA						-0.224				-0.211
						(0.232)				(0.220)
Class Size							0.000424			0.000326
							(0.000348)			(0.000388)
1 = Meeting Twice Per Week								-0.0194		-0.0295
× 1 = Early-morning Class								(0.0172)		(0.0209)
1 = Meeting Twice Per Week									0.0221	0.0309
× 1 = Late-afternoon Class									(0.0217)	(0.0306)
Observations	243	243	243	243	243	243	243	243	243	243
Mean regressor	0.104	0.579	0.0368	0.260	23.59	3.810	71.92	2.181	2.078	
Mean outcome	0.460	0.460	0.460	0.460	0.460	0.460	0.460	0.460	0.460	0.460
Term × Course × Instructor FE	X	X	X	X	X	X	X	X	X	X
Joint significance: p-value										0.208

Notes: The sample consists of 17,477 freshmen enrolled in introductory mathematics courses during their first term across the 2016–2023 entry cohorts. Data are aggregated by terms, course, instructor, and class. The dependent variable is classroom female share. Columns (1)–(9) include one classroom characteristic at a time and report its estimation-sample mean. Early-morning class starts at 8–9:59 a.m; and late-afternoon class starts at 4–6:59 p.m. Column (10) jointly includes White, Asian, Black, and Latino shares, mean age, mean high school GPA, class size, and indicator stating that this class meets twice per week. It also reports the p-value from a joint test that the coefficients on all eight classroom characteristics equal zero. All regressions control for term × course × instructor fixed effects. Standard errors in parentheses are clustered at the same level.

* Statistically significant at the 10 percent level.

** Statistically significant at the 5 percent level.

*** Statistically significant at the 1 percent level.

Table 4: Balance Checks: Actual Allocation and Randomization Inference

	(1) Female	(2) White	(3) Asian	(4) Non-minority	(5) Black	(6) Latino	(7) Minority	(8) International	(9) Age	(10) High School GPA	(11) Declared STEM at Entry
Female share	0.026	0.086	-0.090	-0.005	0.034	-0.022	0.012	-0.036	-0.075	-0.086	0.112
	(0.093)	(0.063)	(0.101)	(0.079)	(0.040)	(0.085)	(0.078)	(0.039)	(0.072)	(0.069)	(0.274)
Mean outcome	0.490	0.108	0.537	0.646	0.042	0.298	0.339	0.028	23.718	3.788	0.743
Observations	17,477	17,477	17,477	17,477	17,477	17,477	17,477	17,477	17,477	17,477	17,477
Term × Course × Instructor FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Two-sided randomization p-value	0.971	0.310	0.458	0.961	0.531	0.875	0.923	0.422	0.486	0.246	0.184

Notes: The sample consists of 17,477 freshmen enrolled in introductory mathematics courses during their first term across the 2016–2023 entry cohorts. Each column reports estimates of Equation 2, with leave-one-out classroom female share as the regressor. Each column also reports a two-sided randomization p-value comparing the observed coefficient with the empirical distribution of coefficients obtained by reestimating the same model under 1,000 simulated classroom allocations. See the text for details. All regressions control for term × course × instructor fixed effects. Standard errors in parentheses are clustered at the same level.

* Statistically significant at the 10 percent level.

** Statistically significant at the 5 percent level.

*** Statistically significant at the 1 percent level.

Table 5: Baseline Results: Gender Peer Effects on Academic Outcomes in Introductory Mathematics Courses

	Completed	Grade above C-
	(1)	(2)
Standardized Female Share × Female	0.043**	0.046**
	(0.020)	(0.022)
Standardized Female Share × Male	0.019	0.032*
	(0.015)	(0.017)
Observations	17,477	17,477
Female mean	0.836	0.802
Male mean	0.832	0.797
P-value (Male = Female)	0.338	0.628
Term × Course × Instructor × Female FE	X	X
Controls	X	X

Notes: The sample consists of freshmen from the 2016–2023 entry cohorts who took introductory mathematics courses during their first term. Each column reports estimates of Equation 1, with standardized leave-one-out classroom female share interacted with female and male indicators. The dependent variable in Column (1) indicates completion of the introductory mathematics course; in Column (2), it indicates earning a letter grade above C- in that course. Each column also reports the p-value from a test of equality between the female and male peer-share coefficients. Female and male outcome means use each regression's estimation sample. All regressions include individual and classroom controls and term × course × instructor × female fixed effects. Standard errors, reported in parentheses, are clustered at the term × course × instructor level.

* Statistically significant at the 10 percent level.

** Statistically significant at the 5 percent level.

*** Statistically significant at the 1 percent level.

Table 6: Robustness Results: Gender Peer Effects on Students' Academic Outcomes in Introductory Mathematics Courses with Controls for Female Share in Other Courses or Classes

	Completed			Grade above C-		
	(1)	(2)	(3)	(4)	(5)	(6)
Standardized Female Share × Female	0.043**	0.042**	0.040*	0.046**	0.045**	0.040*
	(0.020)	(0.020)	(0.021)	(0.022)	(0.021)	(0.022)
Standardized Female Share × Male	0.019	0.018	0.019	0.032*	0.028*	0.033*
	(0.015)	(0.015)	(0.015)	(0.017)	(0.017)	(0.017)
Observations	17,477	17,477	17,477	17,477	17,477	17,477
Female mean	0.836	0.836	0.836	0.802	0.802	0.802
Male mean	0.832	0.832	0.832	0.797	0.797	0.797
P-value (Male = Female)	0.338	0.323	0.414	0.628	0.565	0.810
Term × Course × Instructor × Female FE	X	X	X	X	X	X
Controls	X	X	X	X	X	X
Other-Course Gender Composition × Female		X			X	
Other-Class Gender Composition within the Same Math Course × Female			X			X

Notes: The sample consists of 17,477 freshmen enrolled in introductory mathematics courses during their first term across the 2016–2023 entry cohorts. Each column reports estimates of Equation 1, with standardized leave-one-out classroom female share interacted with female and male indicators. The dependent variable in Columns (1)–(3) indicates completion of the introductory mathematics course; in Columns (4)–(6), it indicates earning a letter grade above C-. Columns (1) and (4) report the baseline specification. Columns (2) and (5) additionally control for average female share in other courses interacted with students' gender indicators. Columns (3) and (6) instead control for average female share in other classes in the same mathematics course interacted with students' gender indicators. Each column also reports the p-value from a test of equality between the female and male peer-share coefficients. All regressions include individual and classroom controls and term × course × instructor × female fixed effects. Standard errors, reported in parentheses, are clustered at the term × course × instructor level.

* Statistically significant at the 10 percent level.

** Statistically significant at the 5 percent level.

*** Statistically significant at the 1 percent level.

Table 7: Heterogeneous Results: by Major Declared at Entry

	Completed (1)	Grade above C- (2)
Panel A. Declared non-STEM major at entry		
Standardized Female Share × Female	-0.046	-0.012
	(0.053)	(0.052)
Standardized Female Share × Male	0.013	0.024
	(0.042)	(0.053)
Observations	4,490	4,490
Female mean	0.822	0.766
Male mean	0.814	0.759
P-value (Male = Female)	0.347	0.644
Panel B. Declared STEM major at entry		
Standardized Female Share × Female	0.054**	0.057**
	(0.021)	(0.023)
Standardized Female Share × Male	0.017	0.033*
	(0.016)	(0.019)
Observations	12,986	12,986
Female mean	0.841	0.817
Male mean	0.837	0.808
P-value (Male = Female)	0.202	0.475
Controls	X	X
Term × Course × Instructor × Female FE	X	X

Notes: The sample consists of 17,477 freshmen enrolled in introductory mathematics courses during their first term across the 2016–2023 entry cohorts. Panels A and B report results for students who declared non-STEM and STEM majors at entry, respectively. Each column reports estimates of Equation 1, with standardized leave-one-out classroom female share interacted with female and male indicators. The dependent variable in Column (1) indicates completion of the introductory mathematics course; in Column (2), it indicates earning a letter grade above C-. Each column also reports the p-value from a test of equality between the female and male peer-share coefficients within the corresponding subsample. Female and male outcome means use each regression's estimation sample. All regressions include individual and classroom controls and term × course × instructor × female fixed effects. Standard errors, reported in parentheses, are clustered at the term × course × instructor level.

* Statistically significant at the 10 percent level.

** Statistically significant at the 5 percent level.

*** Statistically significant at the 1 percent level.

Table 8: Results in the Long Run: Gender Peer Effects in Introductory Mathematics Courses on Long-Run Academic Outcomes

	(1) Time to declare major	(2) Drop out	(3) Graduate within 4 years	(4) STEM degree within 4 years	(5) Non-STEM degree within 4 years
Standardized Female Share × Female	0.044	-0.005	0.002	0.055	-0.054
	(0.334)	(0.026)	(0.031)	(0.051)	(0.039)
Standardized Female Share × Male	0.476**	-0.023	0.023	0.015	0.008
	(0.240)	(0.027)	(0.027)	(0.041)	(0.028)
Observations	11,270	11,270	11,270	11,270	11,270
Female mean	4.433	0.189	0.784	0.425	0.358
Male mean	4.287	0.238	0.715	0.476	0.239
P-value (Male = Female)	0.292	0.236	0.440	0.103	0.013
Term × Course × Instructor × Female FE	X	X	X	X	X
Controls	X	X	X	X	X

Notes: The sample consists of freshmen enrolled in introductory mathematics courses during their first term across the 2016–2020 entry cohorts. Each column reports estimates of Equation 1, with standardized leave-one-out classroom female share interacted with female and male indicators. The dependent variable in Column (1) is time to declare a major, measured in academic terms. Columns (2)–(5) report results for indicators of dropping out, graduating within four academic years, graduating within four academic years with a STEM degree, and graduating within four academic years with a non-STEM degree, respectively. Each column also reports the p-value from a test of equality between the female and male peer-share coefficients. Observations count student-course records, and female and male outcome means use each regression's estimation sample. All regressions include individual and classroom controls and term × course × instructor × female fixed effects, with admissions cohort identifying the term component. Standard errors, reported in parentheses, are clustered at the term × course × instructor level.

* Statistically significant at the 10 percent level.

** Statistically significant at the 5 percent level.

*** Statistically significant at the 1 percent level.

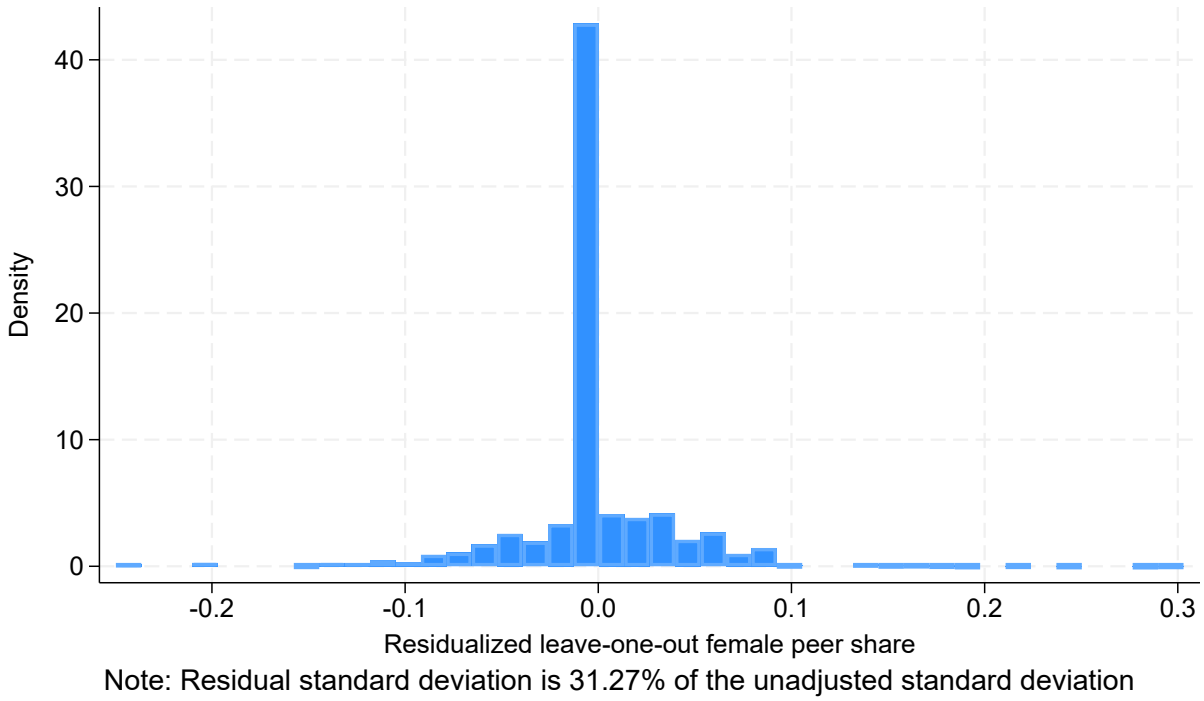


Figure 1: Variations in Residualized Leave-One-Out Female Share in Classes

Notes: The sample consists of 17,477 freshmen enrolled in introductory mathematics courses during their first term across the 2016–2023 entry cohorts. The figure displays a density-scaled histogram of residuals from a regression of classroom female share on term × course × instructor fixed effects.

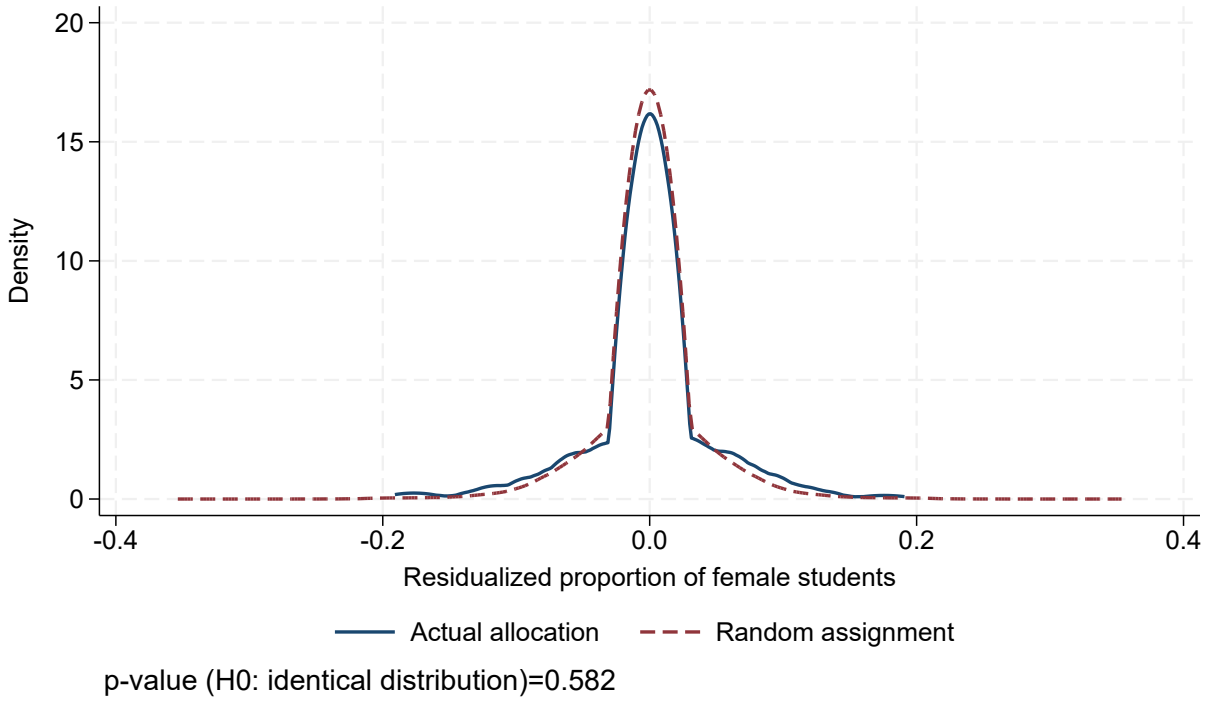


Figure 2: Actual and Simulated Variation in the Proportion of Female Students in Classes

Notes: The sample consists of 17,477 freshmen enrolled in introductory mathematics courses during their first term across the 2016–2023 entry cohorts. The figure plots kernel density estimates of residuals from regressions of classroom female share on admission-cohort × course × instructor fixed effects. The solid line represents observed classroom allocations. The dashed line pools residuals from 1,000 simulations that randomly reassign students to classrooms within each admission-cohort × course × instructor pool, preserving classroom sizes and the total number of female students in each pool. Both density estimates use the same bandwidth.

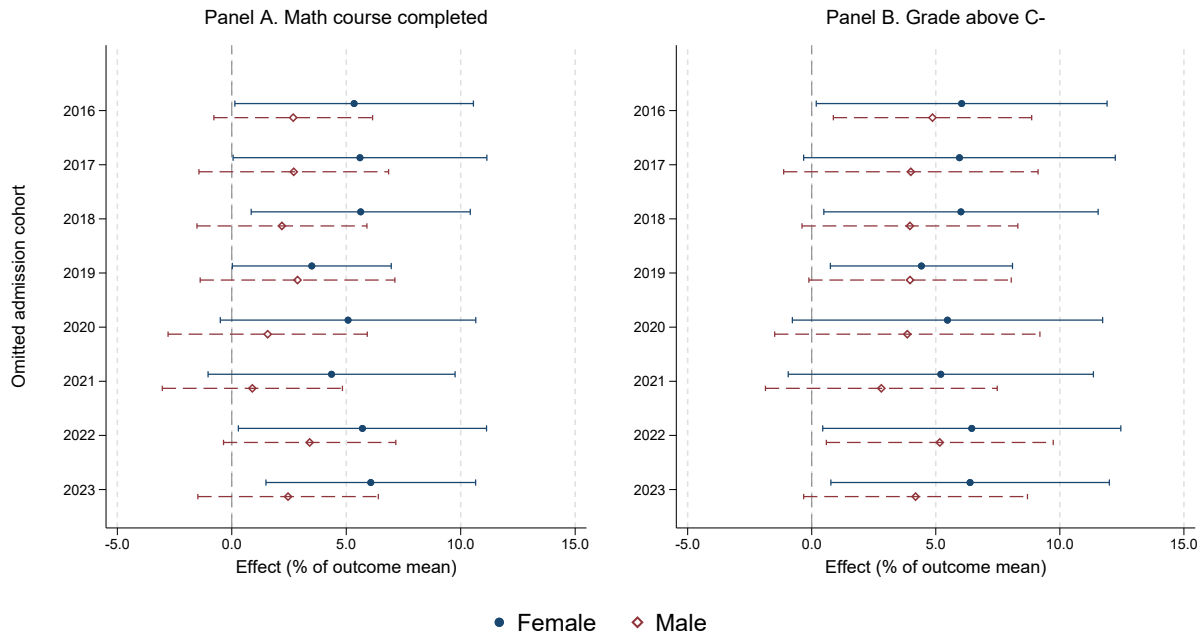


Figure 3: Leave-One-Out Analysis: Gender Peer Effects on Academic Outcomes in Introductory Mathematics Courses with Specific Cohort Excluded

Notes: The sample consists of 17,477 freshmen enrolled in introductory mathematics courses during their first term across the 2016–2023 entry cohorts. Each panel reports estimates of Equation 1, with standardized leave-one-out classroom female share interacted with female and male indicators. The dependent variable in Panel A indicates completion of the introductory mathematics course; in Panel B, it indicates earning a letter grade above C-. Each pair of estimates excludes the admissions cohort indicated on the vertical axis. Blue circles represent female peer-share coefficients, and red hollow diamonds represent male peer-share coefficients. Solid blue and dashed red horizontal lines denote the corresponding 95 percent confidence intervals, based on standard errors clustered at the term × course × instructor level. Estimates and confidence intervals are divided by the corresponding sex-specific estimation-sample outcome mean and multiplied by 100.

Table Appendix

Table A.1: Overview of the Literature on Gender Peer Effects in Higher Education

Study	Regions, institutions, and samples	Variation	Main findings: greater female representation
Oosterbeek and van Ewijk (2014)	Netherlands; University of Amsterdam; first-year economics and business students.	Random assignment across workgroups, conditional on gender.	Men postpone dropping out and perform worse in courses with substantial mathematics content.
Hill (2017)	United States; multiple colleges; freshman cohorts.	Variation across cohorts within colleges.	Men have higher graduation rates.
Booth et al. (2018)	United Kingdom; University of Essex; students in First-year introductory economics courses.	Random assignment to mixed-gender or single-gender classes	Women in all-female classes earn higher grades, are more likely to pass, and have lower subsequent university dropout and a higher probability of obtaining a highly classified degree. Attendance and completion of optional assignments also increase.
Griffith and Main (2019)	United States; college of engineering; first semester.	Random assignment across sections within the course	Both women and men have greater engineering persistence.
Zölitz and Feld (2021)	Netherlands; Maastricht University; School of Business and Economics.	Quasi-random assignment to teaching sections within courses.	Women shift toward female-dominated majors, while men shift toward male-dominated majors. Women experience slower earnings growth.
Bowman et al. (2022)	United States; multiple universities; STEM courses.	Observational variation across courses.	Greater female representation is associated with higher course grades for both women and men.
Bostwick and Weinberg (2022)	United States; public universities in Ohio; doctoral STEM programs.	Variation across cohorts within programs.	Women have higher PhD completion rates and lower rates of early dropout.

Continued on next page

Table A.1: Overview of the Literature on Gender Peer Effects in Higher Education (Cont.)

Study	Regions, institutions, and samples	Variation	Main findings: greater female representation
Zhang (2025)	Singapore; National University of Singapore; Faculty of Arts and Social Sciences.	Quasi-random assignment across tutorial groups within courses.	Both women and men earn higher course grades.
Aguirre et al. (2026)	Chile; multiple universities; STEM and non-STEM programs.	Variation across cohorts within programs in the academic ability of female and male peers.	Higher-ability female peers increase women's STEM graduation and reduce subsequent fertility. Higher-ability male peers reduce women's STEM graduation, employment, and earnings.
Griselda et al. (2026)	Italy; Bocconi University; all departments.	Quasi-random assignment of students to teaching classes within degree programs.	Both women and men improve their academic performance, report lower conformity to masculinity-related attitudes, and perceive less academic interference from worry and anxiety.
Hampole et al. (2026)	United States; a business school; MBA students.	Variation across sections within courses.	Women are more likely to attain senior management positions and move toward female-friendly firms.
Shan (2026)	Europe; a business school; students in an introductory economics course at entry.	Random assignment to mixed-gender or single-gender study groups.	Mixed-gender groups improve grades and well-being and reduce program dropout for both genders.

Table A.2: Summary Statistics: Freshmen Enrolled in All Courses and in Introductory Mathematics Courses in the Entry Term

	Female		Male	
	All Freshmen (1)	Introductory Math (2)	All Freshmen (3)	Introductory Math (4)
White	0.088	0.099	0.11	0.12
	(0.28)	(0.30)	(0.32)	(0.32)
Asian	0.39	0.51	0.50	0.56
	(0.49)	(0.50)	(0.50)	(0.50)
Black	0.061	0.049	0.041	0.035
	(0.24)	(0.22)	(0.20)	(0.18)
Latino	0.45	0.32	0.33	0.27
	(0.50)	(0.47)	(0.47)	(0.44)
International	0.023	0.021	0.044	0.035
	(0.15)	(0.14)	(0.21)	(0.18)
High School GPA	3.77	3.82	3.73	3.76
	(0.28)	(0.28)	(0.31)	(0.30)
Declared STEM at entry	0.42	0.71	0.55	0.77
	(0.49)	(0.45)	(0.50)	(0.42)
Observations	20,266	8,561	16,572	8,916

Notes: This table reports summary statistics for freshmen in their first term during academic years 2016–2023, with all freshmen and those enrolled in introductory mathematics courses. Columns (1) and (2) report statistics for female students in these two samples, respectively; Columns (3) and (4) report the corresponding statistics for male students.

Table A.3: Summary Statistics: By Mathematics Course Group

Course Group	Total Freshmen (1)	Average Female Share (%) (2)	Total Classes (3)	Unique Instructors (4)
Preparatory/precalculus	9,189	55.5 (9.8)	76	31
Standard calculus I	1,648	34.8 (7.6)	32	19
Standard calculus II (advanced)	2,317	35.6 (14.3)	42	24
Standard calculus III (more advanced)	1,000	29.7 (12.4)	30	18
Business calculus	585	50.5 (6.9)	8	2
Life-sciences calculus I	1,323	57.3 (10.6)	25	20
Life-sciences calculus II (advanced)	1,395	54.1 (9.7)	28	19
Other advanced mathematics	20	56.2 (8.8)	2	2
Counts/Average	17,477	46.0 (15.3)	243	89

Notes: The sample consists of 17,477 freshmen enrolled in introductory mathematics courses during their first term across the 2016–2023 entry cohorts. Courses are grouped into preparatory/precalculus, standard calculus, business calculus, life-sciences calculus, and other advanced mathematics based on course numbers and content. Within the standard and life-sciences calculus sequences, Roman numerals reflect the progression indicated by course numbers: higher Roman numerals indicate more advanced courses. Other advanced mathematics includes more difficult courses that fall outside the preceding groups based on their course numbers, content, or prerequisites. Column (1) counts class enrollments among students identified in the first-term introductory mathematics sample; students enrolled in multiple classes contribute to each class count. Column (2) reports the average percentage of female students among freshmen within each class, with the standard deviation across classes in parentheses. Classes receive equal weight, and classes without freshmen are excluded from these calculations. Column (3) counts classes, defined by year, quarter, course, and section. Column (4) counts distinct instructors with nonmissing identifiers. The overall instructor count counts each instructor once across all course groups.

Table A.4: Summary Statistics for Course–Instructor Pools with Multiple Sections

	Course (1)	Course–Instructor (2)	Class (3)
Size	270.78	209.47	100.93
	(129.35)	(110.54)	(46.83)
Freshmen Size	206.51	159.75	76.97
	(148.23)	(128.66)	(55.92)
Female Share (%)	44.96	43.72	44.07
	(13.98)	(14.00)	(15.22)
White Share (%)	10.96	11.23	10.67
	(3.69)	(3.79)	(5.81)
Asian Share (%)	55.56	55.63	55.69
	(14.07)	(13.92)	(15.15)
Black Share (%)	3.63	3.65	3.69
	(2.10)	(2.22)	(3.10)
Latino Share (%)	28.14	27.76	28.29
	(12.46)	(12.38)	(12.88)
International Share (%)	2.79	3.06	3.11
	(3.17)	(3.30)	(4.22)
High School GPA	3.82	3.82	3.81
	(0.10)	(0.11)	(0.11)
# Instructors	1.29	1.00	1.00
	(0.46)	(0.00)	(0.00)
# Classes	2.68	2.08	
	(0.91)	(0.27)	
Counts	41	53	110

Notes: The sample consists of freshmen enrolled in introductory mathematics courses in their entry term during academic years 2016–2023, with sample including only classes whose instructor taught at least two sections of the same course in the same term. Columns (1), (2), and (3) report means at the course, course–instructor, and class levels, respectively. The class is defined as the lecture sections within each course-instructor pairs. Standard deviations are in parentheses. The final row reports the number counted at each level of aggregation.

Table A.5: Summary Statistics: Academic Outcomes of Cohorts 2016–2023 and Cohorts 2016–2020

	Female		Male	
	2016–2023 (1)	2016–2020 (2)	2016–2023 (3)	2016–2020 (4)
Panel A. Short-run Academic Outcomes				
Completed Math Course	0.84	0.84	0.83	0.84
	(0.37)	(0.37)	(0.37)	(0.37)
Grade Higher than C-	0.80	0.80	0.80	0.80
	(0.40)	(0.40)	(0.40)	(0.40)
Share of Courses Completed in Entry Term	0.87	0.88	0.86	0.86
	(0.17)	(0.16)	(0.18)	(0.18)
All entry-term Courses with Grade Higher than C-	0.52	0.53	0.48	0.49
	(0.50)	(0.50)	(0.50)	(0.50)
Entry Term GPA	3.20	3.16	3.20	3.15
	(0.66)	(0.64)	(0.68)	(0.68)
Declared STEM at Entry	0.71	0.71	0.77	0.78
	(0.45)	(0.45)	(0.42)	(0.42)
Panel B. Continuous Academic Outcomes				
Time to Declare A Major (Terms)		4.43		4.29
		(4.13)		(4.19)
Drop Out		0.19		0.24
		(0.39)		(0.43)
Graduating within 4 Academic Years		0.78		0.72
		(0.41)		(0.45)
Graduating within 4 Academic Years with A STEM degree		0.43		0.48
		(0.49)		(0.50)
Graduating within 4 academic Years with A Non-STEM degree		0.36		0.24
		(0.48)		(0.43)
Observations	8,561	5,620	8,916	5,650

Notes: The sample consists of freshmen enrolled in introductory mathematics courses during their first term. Panel A summarizes demographic characteristics and educational backgrounds for both the 2016–2023 and 2016–2020 entry cohorts. Panel B presents longer-run academic outcomes for the 2016–2020 entry cohorts. Columns (1) and (3) report class-level means for female students; Columns (2) and (4) report the corresponding means for male students. Standard deviations are reported in parentheses.

Table A.6: Robustness Results: Comparisons among Different Control Specifications

	Completed			Grade above C-		
	(1)	(2)	(3)	(4)	(5)	(6)
Standardized Female Share × Female	0.040**	0.043**	0.049**	0.040**	0.046**	0.055**
	(0.019)	(0.020)	(0.023)	(0.020)	(0.022)	(0.023)
Standardized Female Share × Male	0.016	0.019	0.011	0.027	0.032*	0.020
	(0.016)	(0.015)	(0.017)	(0.019)	(0.017)	(0.018)
Observations	17,477	17,477	17,477	17,477	17,477	17,477
Female mean	0.836	0.836	0.836	0.802	0.802	0.802
Male mean	0.832	0.832	0.832	0.797	0.797	0.797
P-value (Male = Female)	0.350	0.338	0.219	0.666	0.628	0.250
Term × Course × Instructor × Female FE	X	X	X	X	X	X
Controls		X			X	
Controls × Female			X			X

Notes: The sample consists of freshmen from the 2016–2023 entry cohorts who took introductory mathematics courses during their first term. Each column reports a specification of Equation 1, with standardized leave-one-out classroom female share interacted with female and male indicators. The dependent variable in Columns (1)–(3) indicates completion of the introductory mathematics course; in Columns (4)–(6), it indicates earning a letter grade above C- in that course. Within each outcome, the first specification excludes individual and classroom controls, the second includes these controls with common coefficients across genders, and the third allows their coefficients to differ by student gender. Individual controls include age, race indicators, international status, and high school GPA. Classroom controls include leave-one-out demographic shares, average high school GPA, and a category variable for meeting schedule which interacts an indicator for weekly meeting frequency with class starting time categories. Each column also reports the p-value from a test of equality between the female and male peer-share coefficients. Female and male outcome means use each regression's estimation sample. All regressions include term × course × instructor × female fixed effects, with admissions cohort identifying the term component. Standard errors, reported in parentheses, are clustered at the term × course × instructor level.

* Statistically significant at the 10 percent level.

** Statistically significant at the 5 percent level.

*** Statistically significant at the 1 percent level.

Table A.7: Baseline Results: Entry-Term Academic Outcomes and Higher Grades

	Completion Rate	Share above C-	Grade A- or Better	Grade B- or Better	Entry-term GPA
	(1)	(2)	(3)	(4)	(5)
Standardized Female Share × Female	0.033***	0.035***	0.025	0.044***	0.005
	(0.009)	(0.009)	(0.023)	(0.017)	(0.031)
Standardized Female Share × Male	0.017**	0.026***	0.013	0.048**	0.028
	(0.008)	(0.008)	(0.020)	(0.018)	(0.021)
Observations	17,477	17,477	17,477	17,477	17,477
Female mean	0.873	0.858	0.335	0.657	3.203
Male mean	0.859	0.840	0.349	0.668	3.196
P-value (Male = Female)	0.140	0.348	0.607	0.897	0.359
Term × Course × Instructor × Female FE	X	X	X	X	X
Controls	X	X	X	X	X

Notes: The sample consists of freshmen from the 2016–2023 entry cohorts who took introductory mathematics courses during their first term. Each column reports estimates of Equation 1, with standardized leave-one-out classroom female share interacted with female and male indicators. Columns (1) and (2) report results for the shares of all courses taken in the entry term that were completed and received a letter grade above C-, respectively. The dependent variables in Columns (3) and (4) indicate earning a grade of A- or better and B- or better, respectively, in the introductory mathematics course. Column (5) reports results for entry-term GPA. Each column also reports the p-value from a test of equality between the female and male peer-share coefficients. Female and male outcome means use each regression's estimation sample. All regressions include individual and classroom controls with common coefficients across genders and term × course × instructor × female fixed effects. Standard errors, reported in parentheses, are clustered at the term × course × instructor level.

* Statistically significant at the 10 percent level.

** Statistically significant at the 5 percent level.

*** Statistically significant at the 1 percent level.

Table A.8: Alternative Specifications: Non-linear Gender Peer Effects in Introductory Mathematics Courses

	Completed (1)	Grade above C- (2)
Middle tercile × Female	0.023	0.019
	(0.037)	(0.037)
Top tercile × Female	0.059	0.070*
	(0.041)	(0.041)
Middle tercile × Male	-0.004	-0.012
	(0.018)	(0.014)
Top tercile × Male	0.060**	0.084***
	(0.024)	(0.018)
Observations	17,477	17,477
Female mean	0.836	0.802
Male mean	0.832	0.797
Individual controls	X	X
Classroom controls	X	X
Term × Course × Instructor× Female FE	X	X

Notes: The sample consists of freshmen enrolled in introductory mathematics courses during academic years 2016–2023. Each column reports a nonlinear specification of Equation 1, replacing the continuous peer-share measure with indicators for the middle and highest terciles of standardized leave-one-out classroom female share, interacted with female and male indicators. Terciles are constructed before outcome-specific estimation exclusions. The lowest tercile is the omitted category for each sex. The dependent variable in Column (1) indicates completion of the introductory mathematics course; in Column (2), it indicates earning a letter grade above C-. Female and male outcome means use each regression's estimation sample. All regressions include individual and classroom controls and term × course × instructor × female fixed effects. Standard errors, reported in parentheses, are clustered at the term × course × instructor level.

* Statistically significant at the 10 percent level.

** Statistically significant at the 5 percent level.

*** Statistically significant at the 1 percent level.

Table A.9: Falsification Test: Gender Peer Effects in Courses Outside Introductory Mathematics

	Completed (1)	Grade above C- (2)
Panel A. Basic writing and introductory English		
Standardized Female Share × Female	-0.010*	-0.006
	(0.005)	(0.006)
Standardized Female Share × Male	0.001	0.003
	(0.007)	(0.007)
Observations	18,635	18,635
Female mean	0.852	0.824
Male mean	0.830	0.795
P-value (Male = Female)	0.212	0.298
Panel B. All other courses		
Standardized Female Share × Female	-0.007	0.003
	(0.012)	(0.012)
Standardized Female Share × Male	0.002	0.001
	(0.009)	(0.010)
Observations	74,139	74,139
Female mean	0.834	0.801
Male mean	0.821	0.791
P-value (Male = Female)	0.592	0.929
Controls	X	X
Term × Course × Instructor × Female FE	X	X

Notes: The sample consists of freshmen enrolled in first-term courses during academic years 2016–2023. Panel A reports results for basic writing and introductory English courses. Panel B reports results for all other courses, excluding basic writing, introductory English, and introductory mathematics. Each column reports estimates of Equation 1, with standardized leave-one-out classroom female share interacted with female and male indicators. The dependent variable in Column (1) indicates completion of the course; in Column (2), it indicates earning a letter grade above C-. Each column also reports the p-value from a test of equality between the female and male peer-share coefficients. Observations count student-course records, and female and male outcome means use each regression's estimation sample. All regressions include individual and classroom controls and term × course × instructor × female fixed effects, with admissions cohort identifying the term component. Standard errors, reported in parentheses, are clustered at the term × course × instructor level.

* Statistically significant at the 10 percent level.

** Statistically significant at the 5 percent level.

*** Statistically significant at the 1 percent level.

Table A.10: Baseline Results: by Admission Cohort Coverage

	Completed (1)	Grade above C- (2)
Panel A. Cohorts 2016–2023		
Standardized Female Share × Female	0.043**	0.046**
	(0.020)	(0.022)
Standardized Female Share × Male	0.019	0.032*
	(0.015)	(0.017)
Observations	17,477	17,477
Female mean	0.836	0.802
Male mean	0.832	0.797
P-value (Male = Female)	0.338	0.628
Term × Course × Instructor × Female FE	X	X
Controls	X	X
Panel B. Cohorts 2016–2019		
Standardized Female Share × Female	0.050	0.050
	(0.031)	(0.038)
Standardized Female Share × Male	0.011	0.031
	(0.024)	(0.032)
Observations	9,135	9,135
Female mean	0.836	0.799
Male mean	0.840	0.798
P-value (Male = Female)	0.329	0.712
Term × Course × Instructor × Female FE	X	X
Controls	X	X
Panel C. Cohorts 2016–2020		
Standardized Female Share × Female	0.049**	0.052*
	(0.024)	(0.030)
Standardized Female Share × Male	0.018	0.031
	(0.019)	(0.023)
Observations	11,270	11,270
Female mean	0.837	0.802
Male mean	0.836	0.797
P-value (Male = Female)	0.349	0.600
Term × Course × Instructor × Female FE	X	X
Controls	X	X

Notes: The sample consists of freshmen enrolled in introductory mathematics courses during their first term. Panels A, B, and C report results for the 2016–2023, 2016–2019, and 2016–2020 entry cohorts, respectively. Each column reports estimates of Equation 1, with standardized leave-one-out classroom female share interacted with female and male indicators. The dependent variable in Column (1) indicates completion of the introductory mathematics course; in Column (2), it indicates earning a letter grade above C-. Each column also reports the p-value from a test of equality between the female and male peer-share coefficients. Observations and female and male outcome means use each regression's estimation sample. All regressions include individual and classroom controls and term × course × instructor × female fixed effects, with admissions cohort identifying the term component. Standard errors, reported in parentheses, are clustered at the term × course × instructor level.

* Statistically significant at the 10 percent level.

** Statistically significant at the 5 percent level.

*** Statistically significant at the 1 percent level.

Table A.11: Heterogeneous Results: Gender Peer Gender Effects in Introductory Mathematics Courses by Race

	Completed (1)	Grade above C- (2)
Panel A. White		
Standardized Female Share × Female	-0.047	-0.021
	(0.053)	(0.047)
Standardized Female Share × Male	-0.066	-0.075**
	(0.045)	(0.038)
Observations	1,896	1,896
Female mean	0.837	0.821
Male mean	0.843	0.815
P-value (Male = Female)	0.728	0.260
Panel B. Asian		
Standardized Female Share × Female	0.061**	0.068***
	(0.024)	(0.023)
Standardized Female Share × Male	-0.000	0.019
	(0.017)	(0.020)
Observations	9,386	9,386
Female mean	0.833	0.811
Male mean	0.835	0.808
P-value (Male = Female)	0.040	0.118
Panel C. Minority		
Standardized Female Share × Female	0.030	0.028
	(0.029)	(0.035)
Standardized Female Share × Male	0.058*	0.068**
	(0.032)	(0.032)
Observations	5,930	5,930
Female mean	0.839	0.785
Male mean	0.823	0.768
P-value (Male = Female)	0.556	0.441
Controls	X	X
Term × Course × Instructor × Female FE	X	X

Notes: The sample consists of freshmen enrolled in introductory mathematics courses during academic years 2016–2023. Panels A, B, and C report separate estimates for White, Asian, and minority students, respectively, as identified by the corresponding indicators in the data. Each column reports estimates of Equation 1, with standardized leave-one-out classroom female share interacted with female and male indicators. The dependent variable in Column (1) indicates completion of the introductory mathematics course; in Column (2), it indicates earning a letter grade above C-. Each column also reports the p-value from a test of equality between the female and male peer-share coefficients within the corresponding subgroup. Observations count student-course records, and female and male outcome means use each regression's estimation sample. All regressions include individual and classroom controls and term × course × instructor × female fixed effects. Controls that are constant within a subgroup are absorbed or omitted. Standard errors, reported in parentheses, are clustered at the term × course × instructor level.

* Statistically significant at the 10 percent level.

** Statistically significant at the 5 percent level.

*** Statistically significant at the 1 percent level.

Table A.12: Heterogeneous Results: Gender Peer Gender Effects in Introductory Mathematics Courses by High School GPA

	Completed (1)	Grade above C- (2)
Panel A. Lower high school GPA group		
Standardized Female Share × Female	0.050**	0.038*
	(0.021)	(0.020)
Standardized Female Share × Male	0.006	0.029
	(0.022)	(0.020)
Observations	9,041	9,041
Female mean	0.833	0.790
Male mean	0.829	0.779
P-value (Male = Female)	0.116	0.713
Panel B. Higher high school GPA group		
Standardized Female Share × Female	0.030	0.046
	(0.030)	(0.030)
Standardized Female Share × Male	0.049*	0.047
	(0.027)	(0.031)
Observations	8,436	8,436
Female mean	0.838	0.813
Male mean	0.836	0.819
P-value (Male = Female)	0.588	0.977
Controls	X	X
Term × Course × Instructor × Female FE	X	X

Notes: The sample consists of freshmen enrolled in introductory mathematics courses during academic years 2016–2023. Panels A and B report separate estimates for the lower and higher high school GPA groups, respectively. Groups are defined by dividing the distribution of observed high school GPA within each admissions cohort into two quantile groups, using student-course records. Each column reports estimates of Equation 1, with standardized leave-one-out classroom female share interacted with female and male indicators. The dependent variable in Column (1) indicates completion of the introductory mathematics course; in Column (2), it indicates earning a letter grade above C-. Each column also reports the p-value from a test of equality between the female and male peer-share coefficients within the corresponding subgroup. Observations count student-course records, and female and male outcome means use each regression's estimation sample. All regressions include individual and classroom controls and term × course × instructor × female fixed effects. Standard errors, reported in parentheses, are clustered at the term × course × instructor level.

* Statistically significant at the 10 percent level.

** Statistically significant at the 5 percent level.

*** Statistically significant at the 1 percent level.

Table A.13: Results in the Long Run: Gender Peer Effects in Introductory Mathematics Courses on Long-Run Academic Outcomes by Major Declared at Entry

	Time to declare major (1)	Switch STEM status (2)	Drop out (3)	Graduate within 4 years (4)	STEM degree within 4 years (5)	Non-STEM degree within 4 years (6)
Panel A. Declared non-STEM major at entry						
Standardized Female Share × Female	-0.009	0.014	0.039	-0.025	0.018	-0.043
	(0.671)	(0.032)	(0.055)	(0.062)	(0.030)	(0.058)
Standardized Female Share × Male	1.845**	-0.059	-0.054	0.016	-0.052	0.067
	(0.710)	(0.061)	(0.072)	(0.068)	(0.051)	(0.084)
Observations	2,895	2,895	2,895	2,895	2,895	2,895
Female mean	4.883	0.044	0.172	0.810	0.027	0.782
Male mean	5.621	0.081	0.229	0.742	0.051	0.692
P-value (Male = Female)	0.109	0.198	0.326	0.671	0.260	0.190
Panel B. Declared STEM major at entry						
Standardized Female Share × Female	-0.031	-0.020	-0.037	0.033	0.046	-0.015
	(0.356)	(0.025)	(0.036)	(0.042)	(0.037)	(0.024)
Standardized Female Share × Male	0.236	-0.005	-0.035	0.037	0.037	0.001
	(0.262)	(0.017)	(0.029)	(0.030)	(0.029)	(0.018)
Observations	8,374	8,374	8,374	8,374	8,374	8,374
Female mean	4.246	0.218	0.196	0.773	0.588	0.184
Male mean	3.905	0.138	0.241	0.707	0.597	0.109
P-value (Male = Female)	0.530	0.582	0.920	0.917	0.778	0.572
Controls	X	X	X	X	X	X
Term × Course × Instructor × Female FE	X	X	X	X	X	X

Notes: The sample consists of freshmen enrolled in introductory mathematics courses during their first term from the 2016–2020 entry cohorts. Panels A and B report separate estimates for students who declared non-STEM and STEM majors at entry, respectively. Each column reports estimates of Equation 1, with standardized leave-one-out classroom female share interacted with female and male indicators. The dependent variable in Column (1) is time to declare a major, measured in the units of the source variable. Column (2) indicates switching into STEM in Panel A and switching out of STEM in Panel B. Columns (3)–(6) report results for indicators of dropping out, graduating within four academic years, graduating within four academic years with a STEM degree, and graduating within four academic years with a non-STEM degree, respectively. Each column also reports the p-value from a test of equality between the female and male peer-share coefficients within the corresponding subgroup. Observations count student-course records, and female and male outcome means use each regression's estimation sample. All regressions include individual and classroom controls and term × course × instructor × female fixed effects. Standard errors, reported in parentheses, are clustered at the term × course

* Statistically significant at the 10 percent level.

** Statistically significant at the 5 percent level.

*** Statistically significant at the 1 percent level.

Figure Appendix

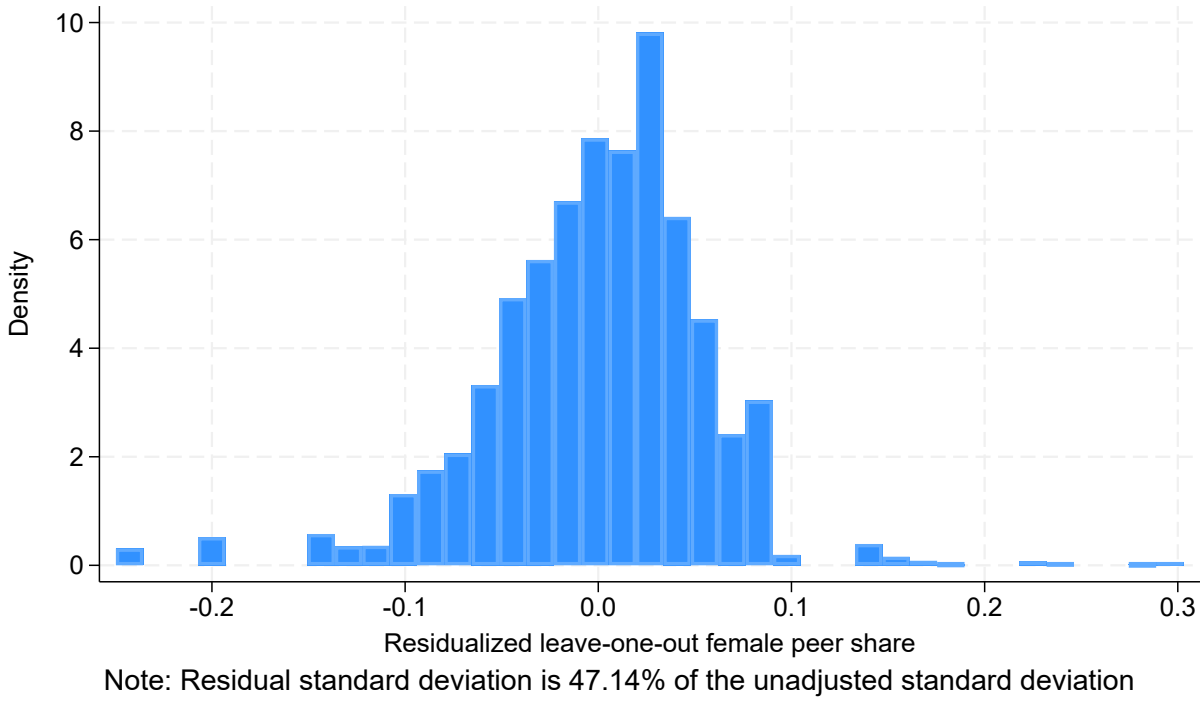


Figure B.1: Variations in Residualized Leave-One-Out Female Share in Classes: Samples of Classes with Instructor Teaching Two or More Classes for the Course-Term

Notes: The sample consists of freshmen enrolled in introductory mathematics courses during academic years 2016–2023, including only classes whose instructor taught at least two sections of the same course in the same term. The figure displays a density-scaled histogram of residuals from a regression of classroom female share on term × course × instructor fixed effects.